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Dihedral influence on lateral–directional dynamic stability on large aspect ratio tailless flying wing aircraft

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Abstract The influence of dihedral layout on lateral–directional dynamic stability of the tailless flying wing aircraft is discussed in this paper. A tailless flying wing aircraft with a large aspect ratio is selected as the object of study, and the dihedral angle along the spanwise sections is divided into three segments. The influence of dihedral layouts is studied. Based on the stability derivatives calculated by the vortex lattice method code, the linearized small-disturbance equations of the lateral modes are used to determine the mode dynamic characteristics. By comparing 7056 configurations with different dihedral angle layouts, two groups of stability optimized dihedral layout concepts are created. Flight quality close to Level 2 requirements is achieved in these optimized concepts without any electric stability augmentation system.

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1. Introduction

Tailless flying wing configuration aircraft have advantages in aerodynamic and structural efficiency because of their simple shape and they have attracted wide interest in both military and civilian fields. Civilian flying wing aircraft include the Boeing X-48,[1,2] the low noise transporter developed by Cranfield University,[3,4] the flying wing aircraft in the studied by the Russian Central Aerohydrodynamic Institute (TsAGI),[5] and the 250-seat flying wing concept in Beihang University.[6] In the field of military aviation, several countries have developed UCAV with tailless configurations, such as the X-45,[7,8] X-47B,[9] nEURO, etc.[10]

An ideal flying wing aircraft blends wings and fuselage into a lifting body without horizontal or vertical stabilizers. Compared to conventional configurations, it is of low yaw stiffness and yaw damp because of the absence of vertical stabilizers. Therefore, tailless aircraft often exhibit poor lateral dynamic stability. Historically, there have been two methods to solve this problem.

The first method originated from the Northrop cooperation, which started its research on tailless aircraft during the

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1940s. Experimental studies on the flight characteristics of tailless aircraft led to the development of the stability augmentation system.[11,12] This technology laid the foundation for the future success of modern tailless aircraft, such as the famous B-2 bomber.[13] Another approach came from the Germany-based Horten brothers, who conducted their research on pure tailless aircraft using the glider testbed during the 1930s.[14] On their successful tailless fighter, the Ho. 229 (Go. 229), an acceptable lateral–directional flight quality was achieved by using an appropriate sweepback angle and tap ratio without any electronic augmentation.

In most pure tailless aircraft developed in recent years, an electric stability augmentation system is used.[15] In contrast, this paper examines a method to increase the lateral dynamic stability on pure tailless aircraft through use of dihedral layout optimization. Considering a global and multidisciplinary audience, the planform and airfoil of each spanwise section of the tailless aircraft are designed to meet applicable aerodynamic and low observability requirements. Assuming a fixed sweepback angle, tap ratio and aspect ratio, only the dihedral angles may be altered to meet the flying quality requirements. Aerodynamic principles indicate that changes in the dihedral angle can alter the angle of attack along the spanwise section while the aircraft is in sideslip, leading to changes in the lift distribution along the spanwise section and changes in the lateral–directional aerodynamic derivatives. However, these changes in lift distribution caused by the dihedral will only work when the aircraft is in sideslip. Thus, the adjustment of the lateral–directional aerodynamic derivatives can be achieved by setting the dihedral angle along the spanwise section, without affecting the lift-drag ratio performance.

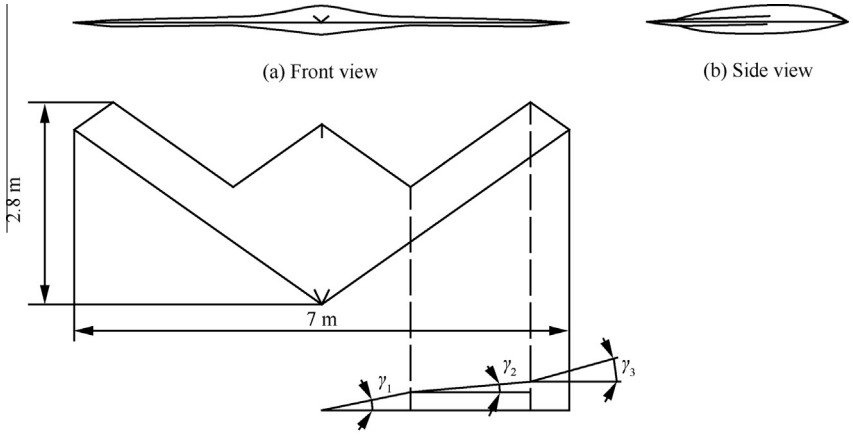
Nickel and Wohlfahrt[14] qualitatively described the effect of the simple shaped dihedral on "Skid Roll" stability, which is a term for lateral static stability, but lateral dynamic stability is decided by further elements of the aircraft. The effect on lateral dynamic stability when using a complex dihedral on a large aspect ratio tailless aircraft was studied in this paper. The vortex lattice method was used to calculate the stability derivatives, and lateral–directional linearization small disturbance equations were used to calculate the characteristics of lateral dynamic stability. Through the calculation and analysis of results from 7056 configurations of dihedral distribution, two groups of lateral stability optimized concepts were created. Flight quality close to Level 2 requirements according to the MIL-8785C[16] was finally achieved using these concepts.

2. Description of study object

A conceptual small tailless unmanned aerial vehicle was selected as the object of study. The initial aircraft parameters were obtained through traditional design methods. The main parameters of the aircraft are as follows: wingspan of 7 m, length of 2.8 m, wing area of 7.2 m^2, and mass of 104 kg for a typical level flight. The planform view of the aircraft is shown in Fig. 1. The moments of inertia are calculated by modeling every component of the aircraft in CATIA. These moments of inertia as related to lateral–directional stability are shown in Table 1.

The distribution of dihedral angles of the aircraft was divided into three sections, as illustrated in Fig 1. To study the influence of different layouts of dihedral angles on lateral–directional stability, the range of dihedral angles considered for the inner section γ_1 and outer section γ_3 span from $-10°$ to $10°$. To avoid affecting the design of landing gears due to low height of wingtips, the range of dihedral angles considered for the middle section γ_2 was restricted between $-5°$ and $10°$. The dihedral angles in the three sections can be changed in intervals of $1°$ individually. Thus, with 21 potential degrees for the inner section, 21 for the outer section, and 16 for the middle section, there were a total of 7056 unique candidate layouts for dihedral angle distribution in the design space.

To study lateral–directional stability of the aircraft at different airspeeds, four airspeeds were selected (20, 30, 40, 50 m/s), and the level flight angle of attack for each airspeed was calculated. Next, the derivatives of lateral motion according to the angle of attack and airspeed were calculated. These aerodynamic coefficients and stability derivatives were calculated using the vortex lattice method.[17–19] This method was chosen for its high computation rate, and its computation accuracy in the range of airspeed in this work was widely validated.[19] In the aerodynamic calculation program, the mean camber surface of the aircraft was divided into n vortex



(c) Vertical view and initial geometric parameters

Fig. 1 Conceptual planform and the definition of initial geometric parameters.

Table 1 Moments of inertia as related to lateral–directional stability.

I_x (kg·m^2)	I_z (kg·m^2)	I_{zx} (kg·m^2)
100.1	124.9	2.2

lattices. The vortex ring model was adopted for the vortex lattices before the trailing edge, and the horseshoe pattern was adopted for the vortex lattices at the trailing edge and the free wakes. The free wakes were along the trajectory of the air stream.

The distribution of the vortex lattice for the whole aircraft is shown in Fig 2. Considering the calculation of the aerodynamic forces during sideslip, the aerodynamic forces applied to every section of the bound vortex line were calculated, and the force application points were set at the center of each vortex section. According to the Kutta–Jukovski theory, the aerodynamic forces on each application point are

$$\boldsymbol{f}_n = \rho_\infty(\boldsymbol{V}_\infty + \boldsymbol{C}_{\mathrm{V}n} \times \boldsymbol{\Gamma})\Gamma_n \tag{1}$$

where $\boldsymbol{f}_n$ denotes the aerodynamic force on certain vortex line section; ρ_∞ and V_∞ denote the density and airspeed of the incoming flow; Γ_n denotes the intensity of the vortex on certain vortex line section; $\boldsymbol{\Gamma}$ denotes the column vector constructed by the intensity of the vortex on each vortex lattice; $\boldsymbol{C}_{\mathrm{V}n}$ denotes the coefficient matrix of the term of induced speed on the center of certain vortex line section. The stability derivatives of the aircraft were then calculated using the differentiation of the aerodynamic forces, as calculated above, for each condition.

The characteristics of the lateral–directional mode were obtained through the eigenvalues of matrix $\boldsymbol{A}_{\mathrm{lat}}$, which was built by substituting the calculated stability derivatives and inertial parameters into the lateral–directional linearized small disturbance equation.[20]

$$\boldsymbol{A}_{\mathrm{lat}} = \begin{bmatrix} \overline{Y}_\beta & \alpha_* + \overline{Y}_p & \overline{Y}_r - 1 & g\cos\theta_*/V_* \\ \overline{L}_\beta & \overline{L}_p & \overline{L}_r & 0 \\ \overline{N}_\beta & \overline{N}_\beta & \overline{N}_\beta & 0 \\ 0 & 1 & \tan\theta_* & 0 \end{bmatrix} \tag{2}$$

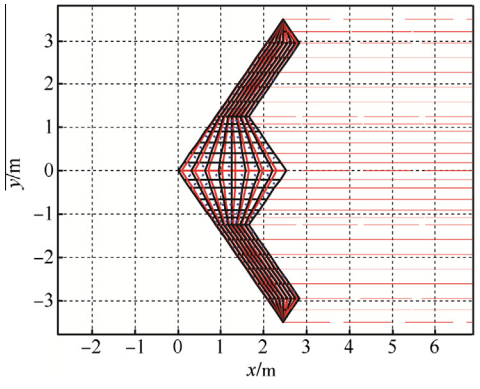


Fig. 2 Vortex lattice layout for aerodynamic calculation.

where

$$\overline{Y}_\beta = \frac{C_{y\beta}q_*S}{mV_*},\ \overline{Y}_p = \frac{C_{yp}q_*S}{mV_*}\cdot\frac{b}{2V_*},\ \overline{Y}_r = \frac{C_{yr}q_*S}{mV_*}\cdot\frac{b}{2V_*}$$

$$N_\beta = C_{n\beta}q_*Sb,\ N_p = C_{np}q_*Sb\frac{b}{2V_*},\ N_r = C_{nr}q_*Sb\frac{b}{2V_*}$$

$$L_\beta = C_{l\beta}q_*Sb,\ L_p = C_{lp}q_*Sb\frac{b}{2V_*},\ L_r = C_{lr}q_*Sb\frac{b}{2V_*}$$

$$\overline{L}_i = \frac{L_i + (I_{zx}/I_z)N_i}{I_x - I_{zx}^2/I_z},\ \overline{N}_i = \frac{N_i + (I_{zx}/I_z)L_i}{I_z - I_{zx}^2/I_x}$$

where $C_{l\beta}$, $C_{n\beta}$, and $C_{y\beta}$ denote the aircraft sideslip derivatives; C_{lp}, C_{np}, and C_{yp} denote the aircraft damping in roll derivatives; C_{lr}, C_{nr}, and C_{yr} denote the aircraft damping in yaw derivatives; I_x, I_z, and I_{zx} denote the aircraft moments of inertia; b denotes the aircraft span; S denotes the wing area; q_* denotes the dynamic pressure; V_* denotes the airspeed; m denotes the mass of aircraft. The dihedral angles of three spanwise segments in the initial design were zero. The variation of the Dutch roll mode characteristics relevant to airspeed is illustrated in Fig 3. These characteristics declined with the increase of airspeed, as shown in Fig 3. When the airspeed was above

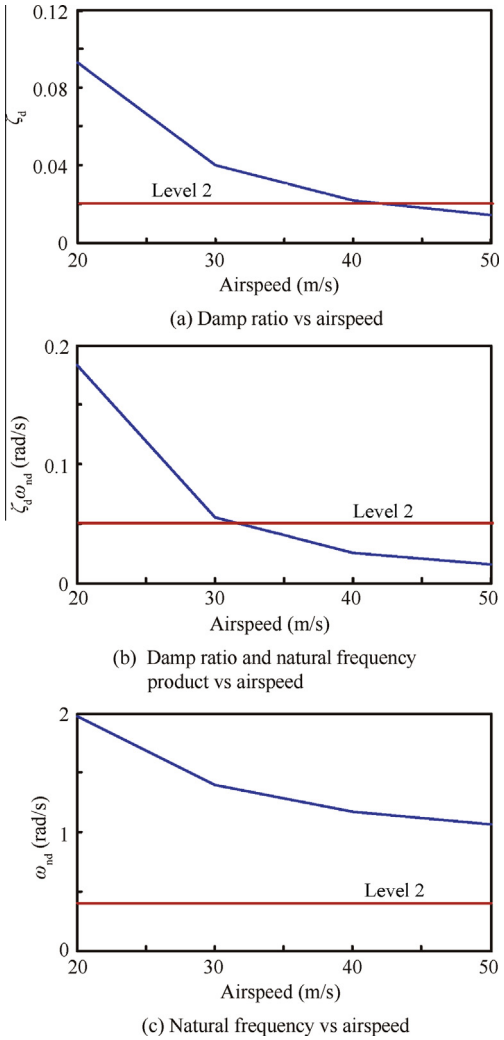


Fig. 3 Variation of Dutch roll mode characters relevant to airspeed.

32 m/s, the product of the damp ratio and the natural frequency will be below the requirement of the Level 2 flight qualities in MIL-8787C. This was mainly caused by the decline of damp ratio with the increase of airspeed (Fig. 3(a)). Thus, an optimized configuration with moderate dihedral layout will be desired, which needs to achieve approximately the flight quality of Level 2 in the entire range of needed airspeeds.

3. Influence of dihedral layout on characteristics of Dutch roll mode

The natural frequency of the Dutch roll mode for each configuration declined with the increase of airspeed. When the airspeed was between 20 m/s and 30 m/s, the natural frequency for each configuration was above 1 rad/s, which met the flight quality requirements for Level 1. When the airspeed reached 30 m/s, the natural frequencies of some configurations declined below 0.7 rad/s, but a considerable number of those were still above 1 rad/s. When the airspeed reached 40 m/s, the natural frequencies of a subset of configurations ceased to decline and instead started to increase slightly, but the majority of them declined even further. A subset of configurations had severe deteriorations, with the natural frequency declining to two real roots; this finding indicated the disappearance of Dutch roll mode. In total, 121 configurations and 2% of the entire design space were in this category. Overall, there were 6781 configurations that met the Level 2 requirements for natural frequency when the airspeed was between 20 m/s and 40 m/s, which accounted for approximately 96% of the design space.

With the increase of airspeed, the damp ratio of each configuration also declined with a gradually slowing rate. When the airspeed was 20 m/s, the damp ratio of each configuration was positive and the vast majority of them were above 0.2, which was in line with the flight quality requirements of Level 2. When the airspeed increased to 30 m/s, a portion of them degraded into the negative, with the overall trend being one of decline. More negative dampers appeared as the airspeed increases further. When the airspeed was in the 20–40 m/s range, the number of configurations with negative damper and degradation of Dutch roll mode reached 3462, which was approximately half of the design space. This result indicated that the divergence of Dutch roll mode was very likely to appear if the lateral–directional configuration parameters of tailless aircraft are designed improperly. On the other hand, there were 2476 configurations that meet the flight quality requirements of Level 2; this number accounted for approximately 35% of the design space.

In addition to natural frequency and damp ratio, the product of these two parameters is itself a requirement set by MIL-8785C as one of the flight quality judging criteria of Dutch roll mode. Its trend of variation with the increase of airspeed combines those of natural frequency and damp ratio: below the speed of 30 m/s, the products of those for all configurations were positive, which indicated convergence of Dutch roll mode. When the airspeed reached 30 m/s and above, the product of natural frequency and damp ratio declined because of individual declines in each factor. When the speed reached 40 m/s, 1382 configurations were still above the requirements for Level 2, which accounted for approximately 20% of the design space.

Considering the three main parameters above, the characteristics of Dutch roll mode on the tailless aircraft studied in this work are shown as follows:

(1) The natural frequencies of Dutch roll mode for most configurations could meet the flight quality requirements of Level 2 within the range of airspeeds they were designed for.
(2) The appearance of negative damper and the degradation of eigenvalues to two real roots existed in approximately half of the configurations, and their Dutch roll mode tends toward divergence.
(3) All configurations demonstrated the trend that the stability of Dutch roll mode deteriorates with the increase of airspeed.

To obtain a configuration with optimized characteristics of Dutch roll mode, the design space was divided along a boundary defined by 95% of the requirements for Level 2 flight quality, and the remaining configurations shown in Table 2 were produced. The 95% of the requirements for Level 2 flight quality requires natural frequency of above 0.38 rad/s, the damp ratio of above 0.019, and the product of natural frequency with damp ratio of above 0.0475 rad/s. For the configurations of dihedral layout in Table 2, the natural frequency, damp ratio, and their product were all above the 95% boundary for the requirements of Level 2 flight qualities covering airspeeds from 20 m/s to 40 m/s. There are 1354 configurations in Table 2, which account for 19% of the design space. The configurations whose dihedral angles in the middle section (γ_2) were more than 2° were neglected in Table 2, as the Dutch roll mode characteristics for this type of configuration did not meet the requirements. The introduction of Table 2 is as follows: taking the first column of Table 2 for example, the dihedral angle γ_2 in middle section corresponding to this column is −5°, and the dihedral angle in inner section γ_1 corresponding to row one is −10°. Combining the value 7°–10° in the first row and first column, it indicates that when the dihedral angle γ_2 in middle section is −5° and the dihedral angle in inner section γ_1 is −10°, the Dutch roll mode characters could meet 95% of the requirements for Level 2 flight quality when the dihedral angle in outer section γ_3 is in 7°–10°.

4. Influence of dihedral angle on spiral mode and rolling mode

It is specified in MIL-8785C that the criteria for spiral mode is the lower bound of the time to double the amplitude of disturbance on a roll angle of less than 20°. The calculation results suggest that the time to double the amplitude for the divergence of spiral mode presented several trends for variation with the increase of airspeed. Below the airspeed of 40 m/s, the time to double the amplitude for the divergence of spiral mode for most configurations (7016) was larger than those when the airspeed was below 20 m/s. At the same time, spiral mode for a considerable portion of configurations in the design space trended toward convergence when the airspeed was below 30 m/s. Setting the specification of the flight quality for Level 2 on the time to double the amplitude for the divergence of spiral mode as the reference, which was less than 7.6 s (including the convergence of spiral mode), and dividing the design space with the boundary of 95% of the requirement above. Table 3 shows the distribution of dihedral angles of the configurations that failed to meet the requirement, and its form is as same as Table 2. Thus, there are 1998 configurations, which account for approximately 28% of the design space.

Table 2 Dihedral layouts meet 95% of Level 2 requirements in Dutch roll mode.

γ_1 (°)	γ_3 (°)							
	$\gamma_2 = -5°$	$\gamma_2 = -4°$	$\gamma_2 = -3°$	$\gamma_2 = -2°$	$\gamma_2 = -1°$	$\gamma_2 = 0°$	$\gamma_2 = 1°$	$\gamma_2 = 2°$
−10	7 to 10	7 to 10	8 to 10	8 to 10		−10	−10 to −8	−10
−9	4 to 10	4 to 10	4 to 10	−10, 4 to 10	−10 to −7	−10 to −5	−10 to −8	
−8	1 to 10	−10 to −9, 0 to 10	−10 to 10	−10 to 10	−10 to 0	−10 to −5	−10 to −9	
−7	−10 to 10	−10 to 10	−10 to 10	−10 to 10	−10 to −2	−10 to −6	−10	
−6	−10 to 10	−10 to 10	−10 to 10	−10 to 4	−10 to −3	−10 to −8		
−5	−10 to 10	−10 to 10	−10 to 10	−10 to 1	−10 to −5	−10 to −9		
−4	−10 to 10	−10 to 10	−10 to 10	−10 to −1	−10 to −6	−10		
−3	−10 to 10	−10 to 10	−10 to 10	−10 to −2	−10 to −7	−10		
−2	−10 to 10	−10 to 10	−10 to 4	−10 to −3	−10 to −8			
−1	−10 to 10	−10 to 10	−10 to 2	−10 to −4	−10 to −8			
0	−10 to 10	−10 to 10	−10 to 1	−10 to −5	−10 to −9			
1	−10 to 10	−10 to 10	−10 to 0	−10 to −6	−10 to −9			
2	−10 to 10	−10 to 10	−10 to −1	−10 to −6	−10			
3	−10 to 10	−10 to 10	−10 to −1	−10 to −6	−10			
4	−10 to 10	−10 to 10	−10 to −1	−10 to −6	−10			
5	−10 to 10	−10 to 10	−10 to −2	−10 to −6	−10			
6	−10 to 10	−10 to 10	−10 to −1	−10 to −6	−10			
7	−10 to 10	−10 to 10	−10 to −1	−10 to −6	−10			
8	−10 to 10	−10 to 10	−10 to −1	−10 to −6	−10			
9	−10 to 10	−10 to 10	−10 to 0	−10 to −6	−10			
10	−10 to 10	−10 to 10	−10 to 1	−10 to −5	−10 to −9			

Table 3 Dihedral layouts failing to meet 95% of Level 2 requirements in spiral mode.

γ_1 (°)	γ_3 (°)							
	$\gamma_2 = -5°$	$\gamma_2 = -4°$	$\gamma_2 = -3°$	$\gamma_2 = -2°$	$\gamma_2 = -1°$	$\gamma_2 = 0°$	$\gamma_2 = 1°$	$\gamma_2 = 2°$
−10	−10 to 10	−10 to 10	−10 to 10	−10 to 9	−10 to 4	−10 to 2	−10 to 2	−10
−9	−10 to 10	−10 to 10	−10 to 10	−10 to 9	−10 to 4	−10 to −1	−10 to −6	
−8	−10 to 10	−10 to 10	−10 to 10	−10 to 9	−10 to 3	−10 to −2	−10 to −6	
−7	−10 to 10	−10 to 10	−10 to 10	−10 to 8	−10 to 3	−10 to −2	−10 to −7	
−6	−10 to 10	−10 to 10	−10 to 10	−10 to 8	−10 to 3	−10 to −2	−10 to −7	
−5	−10 to 10	−10 to 10	−10 to 10	−10 to 8	−10 to 2	−10 to −3	−10 to −7	
−4	−10 to 10	−10 to 10	−10 to 10	−10 to 7	−10 to 2	−10 to −3	−10 to −8	
−3	−10 to 10	−10 to 10	−10 to 10	−10 to 7	−10 to 2	−10 to −3	−10 to −8	
−2	−10 to 10	−10 to 10	−10 to 10	−10 to 6	−10 to 1	−10 to −4	−10 to −8	
−1	−10 to 10	−10 to 10	−10 to 10	−10 to 6	−10 to 1	−10 to −4	−10 to −9	
0	−10 to 10	−10 to 10	−10 to 10	−10 to 5	−10 to 0	−10 to −5	−10 to −9	
1	−10 to 10	−10 to 10	−10 to 10	−10 to 5	−10 to 0	−10 to −5	−10	
2	−10 to 10	−10 to 10	−10 to 9	−10 to 4	−10 to −1	−10 to −6		
3	−10 to 10	−10 to 10	−10 to 9	−10 to 3	−10 to −2	−10 to −7		
4	−10 to 10	−10 to 10	−10 to 8	−10 to 3	−10 to −2	−10 to −7		
5	−10 to 10	−10 to 10	−10 to 7	−10 to 2	−10 to −3	−10 to −8		
6	−10 to 10	−10 to 10	−10 to 6	−10 to 1	−10 to −4	−10 to −9		
7	−10 to 10	−10 to 10	−10 to 5	−10 to 0	−10 to −5	−10 to −9		
8	−10 to 10	−10 to 10	−10 to 5	−10 to −1	−10 to −6	−10		
9	−10 to 10	−10 to 9	−10 to 4	−10 to −2	−10 to −6			
10	−10 to 10	−10 to 8	−10 to 3	−10 to −2	−10 to −7			

As for rolling mode, the trend of variation of the time constant with the increase of airspeed was one of the monotonous decreases in the range of 20–40 m/s. For certain airspeeds, the variation of the time constant of rolling mode between each configuration was less than 8% and was much less than 1.0 s, which was specified in MIL-8785C for Level 1 flight quality. Thus, it could be considered that all distributions of dihedral angle in the design space would not have an effect on rolling mode that influences the flight safety.

5. Overall analysis of influence of dihedral angle

Rolling mode of all configurations in the design space met the flight quality requirements of Level 1, so the flight quality of the aircraft studied in this work was decided mainly by their Dutch roll mode and spiral mode. If the feasible region was divided by the boundary of 95% of the requirement of the Level 2 flight qualities, through the calculation and analysis of the design space, it could be observed that there is a consid-

Table 4 Dihedral layout to meet 95% of Level 2 requirements in lateral stability.

Parameter		Group 1			Group 2				
Dihedral layouts	γ_1 (°)	9	10	10	−10	−9	−8	−7	−7
	γ_2 (°)	−4	−4	−4	−2	−2	−2	−2	−2
	γ_3 (°)	10	9	10	10	10	10	9	10
V = 20 m/s	ω_{nd}	2.01	2.00	2.01	1.89	1.90	1.91	1.92	1.93
	ζ_d	0.11	0.11	0.11	0.12	0.12	0.12	0.12	0.12
	$\zeta_d\omega_{nd}$	0.22	0.22	0.21	0.23	0.23	0.23	0.22	0.22
	T_{da}	7.67	7.67	7.79	7.65	7.68	7.72	7.63	7.76
V = 30 m/s	ω_{nd}	1.47	1.46	1.47	1.24	1.27	1.30	1.31	1.32
	ζ_d	0.06	0.06	0.06	0.08	0.07	0.07	0.07	0.07
	$\zeta_d\omega_{nd}$	0.09	0.09	0.09	0.10	0.10	0.09	0.09	0.09
	T_{da}	22.00	22.34	22.83	23.55	23.36	23.24	22.51	23.17
V = 40 m/s	ω_{nd}	1.30	1.27	1.29	0.88	0.95	1.00	1.03	1.05
	ζ_d	0.05	0.05	0.05	0.08	0.07	0.06	0.06	0.05
	$\zeta_d\omega_{nd}$	0.06	0.06	0.06	0.07	0.06	0.06	0.06	0.06
	T_{da}	45.15	46.80	47.87	56.45	53.37	51.37	48.01	50.00
V = 50 m/s	ω_{nd}	1.27	1.22	1.25	0.58	0.72	0.82	0.87	0.91
	ζ_d	0.04	0.04	0.04	0.10	0.08	0.06	0.06	0.05
	$\zeta_d\omega_{nd}$	0.05	0.05	0.05	0.06	0.06	0.05	0.05	0.05
	T_{da}	79.10	83.60	85.51	143.45	108.31	96.49	85.30	90.44

erable area of overlap between the feasible Region A, which met the requirements for Dutch roll mode, and the infeasible Region B, which failed to meet the requirements for spiral mode. Removing the overlapping region from Region A, a group of configurations with optimized lateral–directional stability could be obtained, which is shown in Table 4.

The optimal configurations with its lateral–directional characteristics are shown in Table 4. The symbols ω_{nd}, ζ_d, and T_{da} are the natural frequency of Dutch roll mode, damp ratio of Dutch roll mode and the time to double the amplitude of spiral mode respectively. The dihedral layouts in Table 4 can be divided into two groups, with either positive or negative dihedral angles in the inner section (γ_1). The distribution of dihedral angles in Group 1 is positive in the inner section, negative in the middle section, and positive in the outer section. Those in Group 2 are negative in the inner and middle sections, and positive in the outer section. For the characteristics of Dutch roll mode, the configurations in Group 1 provide better quality, with natural frequencies of 1.0 rad/s or above for the entire range of considered airspeeds, than those in Group 2. The configurations in Group 2 have some benefit in damp ratio, which is above 20% larger than those of Group 1 in the same airspeed. The product of natural frequency and damp ratio for each airspeed are similar for both Group 1 and Group 2. For the characteristics of spiral mode, the results are comparable at low airspeeds. Group 2 shows few benefits over Group 1 at high airspeeds; the few benefits are not particularly useful as the time to double the amplitude at these airspeeds is already better than the Level 1 flight quality requirements.

The configurations in Group 1 are similar to the "Weltensegler-crank",[14] and it is stated in Karl Nickel and Michael Wohlfahrt's book that many model flying wing aircraft overcame the problem of lateral stability with this configuration. As for Group 2 of configurations, a miniature model flying wing was made to conduct the flight test for qualitative research, with a dihedral layout very similar to that of Group 2 (Fig. 4). Its dihedral angles from inner section to outer section are −6°, −2°, and 10°. The experimental results suggest that this aircraft is of good lateral–directional flight quality. Its handling quality was even better than the aircraft with conventional configuration tested at the same time, especially during takeoff or landing with crosswind.



Fig. 4 Miniature model flying wing.

6. Conclusion

(1) Poor lateral dynamic stability is shown in the original concept with 0° dihedral layout along the entire spanwise sections. Its lateral flight quality will decline with the increase of the airspeed, and finally deteriorate below the flight quality requirements of Level 2.
(2) Approximately 20% of configurations in the design space have met the Level 2 flight quality requirements for Dutch roll mode. The reason for poor Dutch roll stability lies in the lack of adequate damp ratio, especially because damp ratio will decrease further as airspeed increases.
(3) The majority of configurations in the design space exhibit the trend that the stability of spiral mode increases with higher airspeed. If the feasible region is divided by the boundary of 95% of the flight quality requirements of

Level 2, there is a considerable area of overlap between the feasible region of Dutch roll mode and the infeasible region of spiral mode.

(4) All distributions of dihedral angle in the design space have no significant influence on rolling mode characteristics.

(5) Two groups of optimized dihedral layouts are provided by considering all of the lateral mode characteristics. These layouts can maintain flight quality close to Level 2 requirements when the airspeed is between 20 m/s and 40 m/s.

(6) For the aircraft with higher airspeed and larger size compared to that in this work, the influence of its dihedral layout to the lateral–directional dynamic stability requires further research due to the difference in the moment of inertial and wing loading.

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